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Supplementary Materials

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Recyclable, Strong Thermosets and Organogels via Paraformaldehyde Condensation with Diamines

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Nitrogen-based thermoset polymers have many industrial applications (for example, in composites), but are difficult to recycle or rework. We report a simple one-pot, low-temperature polycondensation between paraformaldehyde and 4,4'-oxydianiline (ODA) that forms hemiaminal dynamic covalent networks (HDCNs), which can further cyclize at high temperatures, producing poly(hexahydrotriazine)s (PHTs). Both materials are strong thermosetting polymers, and the PHTs exhibited very high Young's moduli (up to ~14.0 gigapascals and up to 20 gigapascals when reinforced with surface-treated carbon nanotubes), excellent solvent resistance, and resistance to environmental stress cracking. However, both HDCNs and PHTs could be digested at low pH (<2) to recover the bisaniline monomers. By simply using different diamine monomers, the HDCN- and PHT-forming reactions afford extremely versatile materials platforms. For example, when poly(ethylene glycol) (PEG) diamine monomers were used to form HDCNs, elastic organogels formed that exhibited self-healing properties.

Nitrogen-containing thermosets, such as poly(amide)s, poly(imide)s, and poly(benzimidazole)s, are used to manufacture adhesives, coatings, foams, automotive and aerospace parts, and electronic devices owing to their high mechanical strength and durability. Despite their widespread commercial use and practical utility, there has been little recent success in the synthesis and design of new high-performance thermosets because of their at-times challenging production. The development of new synthetic platforms that allow for structural modification to impart a variety of useful materials properties, such as high strength or self-healing, is therefore of considerable interest. Furthermore, all known thermosetting polymers are difficult to recycle because they cannot be remolded once cured and thermally decompose upon heating to

high temperatures. As a result, a thermoset that can be inherently depolymerized would afford a useful route for thermoset recyclability and redeployment.

We have developed two new related classes of thermosets. The first class, hemiaminal dynamic covalent networks (HDCNs; Fig. 1B, 1.5) (1), was prepared by polymerizing monomers containing two –NH₂ units, such as 4,4'-oxydianiline (ODA, 1.4) or diamine-terminated poly(ethylene glycol) (PEG, 4.2), with paraformaldehyde at low (~50°C) temperatures. HDCNs exhibited highly versatile properties depending on the diamine monomer used during the polymerization, and properties range from high-strength, chemically robust materials to self-healing organogels that exhibited chemical reversibility in physiological pH regimes. Furthermore, a specific class of HDCNs prepared from ODA underwent a chemical rearrangement when heated to ~200°C and formed the second class of materials, poly(1,3,5-hexahydro-1,3,5-triazines) (PHTs; Fig. 1B, 1.6), which are highly cross-linked polymer networks.

This work was partially inspired by the synthesis of small-molecule 1,3,5-hexahydro-1,3,5-triazines (HTs; Fig. 1A, 1.3) (2–7), which involves the condensation of amines with formaldehyde proceeding through a hemiaminal (HA) intermedi-

ate (8). To probe the viability of HT formation as a polymer-forming reaction, we studied the trimerization of monofunctional, electron-rich *N,N*-dimethyl-*p*-phenyleneamine with paraformaldehyde as a small-molecule model system by ¹H nuclear magnetic resonance (¹H NMR) analysis (Fig. 1A). HAs formed within 30 min at 50°C, and at 185°C, the corresponding HT was cleanly synthesized (>98% conversion) within 10 min in deuterated dimethyl sulfoxide (*d*₆-DMSO, fig. S14). Thus, ¹H NMR analysis confirmed successful conditions for both HA and HT formation in the model system at different temperatures. Motivated by these results, we explored the condensation reaction of the electron-rich bisaniline monomer, ODA, with paraformaldehyde as a possible route to synthesize HDCNs and PHTs. Experiments were performed to determine if the reaction between ODA and paraformaldehyde could form the requisite HA intermediate, which could then undergo cyclotrimerization.

The condensation of ODA with paraformaldehyde in *N*-methyl pyrrolidone (NMP, 50°C for 30 min) yielded an intractable polymer powder that was characterized by solid-state ¹³C NMR. Signals corresponding to aliphatic and aromatic carbons were consistent with reported values for HT, but line broadening inherent in the method made it difficult to differentiate between HDCN and PHT polymers (figs. S3 and S4). Thus, we performed solid-state infrared (IR) measurements on the polymer film after casting at 50°C (Fig. 1C). The sharp signal representative of the NH₂ stretch in ODA was no longer present after heating (fig. S12), and a new set of C–H IR stretching frequencies had been generated, indicative of sp²-hybridized methylene groups incorporated into the structure. In addition to these expected changes in IR signals, a broad signal at ~3300 cm⁻¹, characteristic of an OH stretching signal expected for an HDCN, was measured. Also observed by IR was the carbonyl stretching frequency corresponding to NMP, which presumably remained bound to the polymer (9). However, upon ramping the cure temperature to 200°C, the solid-state IR spectrum of the resulting material lacked the broad signal at ~3300 cm⁻¹, suggesting that the OH group was no longer present after heating. In addition, ¹H NMR and gel permeation chromatography (GPC) analysis of a soluble polymer was consistent with the construction of an ODA HDCN at 50°C, whereas at 200°C a PHT was formed (fig. S16).

These structural results were supported with calculations using density functional theory (DFT)

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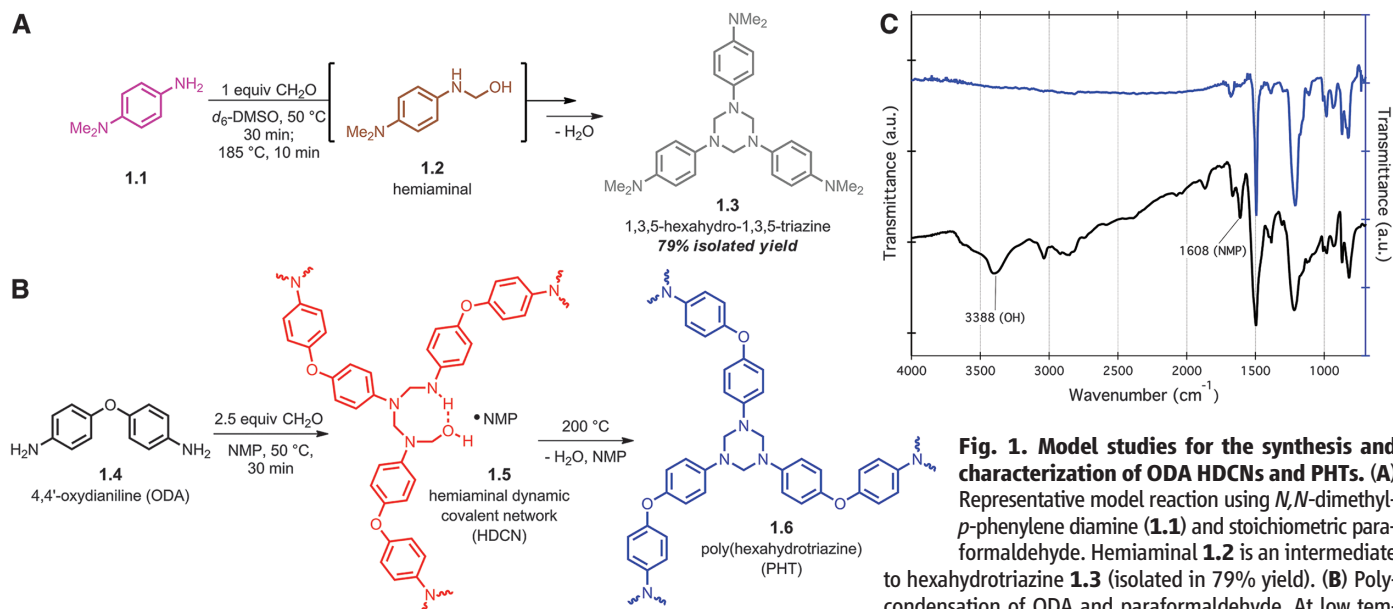


Fig. 1. Model studies for the synthesis and characterization of ODA HDCNs and PHTs. (A) Representative model reaction using *N,N*-dimethyl-*p*-phenylene diamine (**1.1**) and stoichiometric paraformaldehyde. Hemiaminal **1.2** is an intermediate to hexahydrotriazine **1.3** (isolated in 79% yield). (B) Polycondensation of ODA and paraformaldehyde. At low temperatures, an HDCN formed, plasticized with ~27% NMP. (C) Solid-state IR spectra for ODA films cured at 50 °C (black trace) exhibited a broad signal at ~3300 cm^{-1} compared to a film cured at 200 °C (blue trace), which lacked the signal corresponding to the OH stretch. The hydrogen-bonded NMP-related carbonyl stretching frequency is not observed after a 200 °C curing process. (a.u., arbitrary units)

and water. At higher temperatures, the PHT formed. (C) Solid-state IR spectra for ODA films cured at 50 °C (black trace) exhibited a broad signal at ~3300 cm^{-1} compared to a film cured at 200 °C (blue trace), which lacked the signal corresponding to the OH stretch. The hydrogen-bonded NMP-related carbonyl stretching frequency is not observed after a 200 °C curing process. (a.u., arbitrary units)

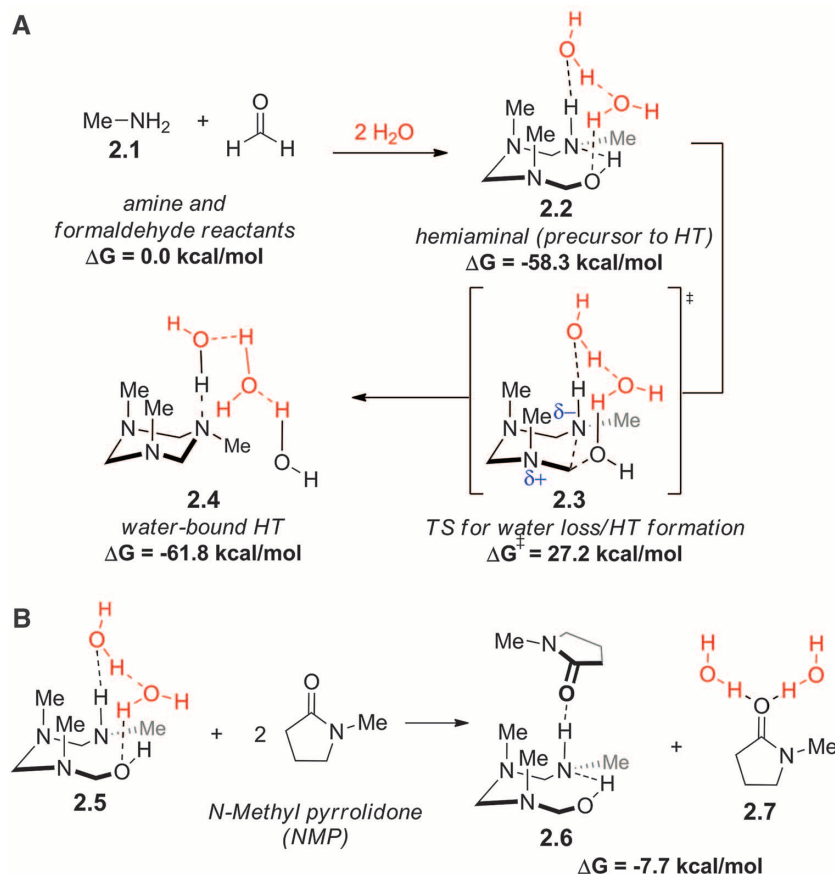


Fig. 2. Mechanism and DFT studies of the water-promoted formation of HT from the reaction of methylamine with formaldehyde. (A) Structures and free energies, in kcal/mol, for the water-promoted conversion of methylamine and formaldehyde into the HA and eventual formation of the HT. The transition structure and barrier for HT formation is shown. (B) Displacement of water from the water-bound HA by NMP and formation of an NMP-bound HA precursor to HT. TS, transition state.

(10–13) on the water-promoted reaction of methylamine (**2.1**) with formaldehyde, which indicated that the formation of HA (**2.2**) was exergonic by about 58 kcal/mol (Fig. 2A). Moreover, the barrier for water loss from this intermediate, leading to cyclization and spontaneous HT formation, was >27 kcal/mol. DFT calculations also predicted that the HA was further stabilized by NMP. The displacement of bound water molecules from the HA precursor (**2.5**) to the HT by NMP solvent produced a complex (**2.6**) in which NMP is hydrogen-bonded to the HA precursor; this process was calculated to be exergonic by 7.7 kcal/mol (Fig. 2B and fig. S30). Thus, the total free energy required for HT formation after exposure to NMP must be a combination of the free energy required to displace the solvent and that required to eliminate water, which was 34.9 kcal/mol. Together, these calculations suggested that the polycondensation between ODA and formaldehyde at low temperature resulted in the formation of a solvent-stabilized intractable PHT precursor (i.e., the NMP/water-bound HDCN), and that PHT formation will only occur at higher temperatures at which the HDCN can be converted to the water-bound PHT through a high-energy transition state. A first-order glass transition temperature (T_g) was observed in the differential scanning calorimetric (DSC) measurements (193 °C, fig. S9 and table S2), and swelling measurements of the PHTs in NMP showed no solvent uptake, substantiating a highly cross-linked network.

These unreinforced polymers exhibited high Young's moduli [(E) , ~14.0 GPa by nanoindentation (*14*), 10.3 GPa by Instron analysis], placing them among the strongest amorphous thermosets known (Fig. 3B and figs. S25 and S26). Carbon

nanotube (CNT)-reinforced composites (**4.1**) were also studied to determine the effect of the matrix resin on the ODA PHT modulus. Carboxylic acid-functionalized CNTs were prepared and loaded in-

to ODA PHT monomer solutions in 2 and 5 weight percent (wt %). Films were cured according to a temperature-ramping procedure to form the ODA PHT (details are provided in the supplementary

materials). Although CNT loading conditions were not optimized, the Young's modulus improved from ~14.0 GPa in the unreinforced film to ~20.0 GPa in the CNT dispersed films, as measured with nanoindentation (Figs. 3B and 4). The Young's modulus value obtained for the ODA PHT-CNT composite approaches moduli for epoxy, polyester, and nylon composites. Although PHTs and their composites show distinctive mechanical properties and exceptional stability to organic solvents and solutions with pH >3, they could be easily depolymerized at pH <2, allowing the recovery of monomers for reuse. The Young's moduli for the ODA HDCN was determined to be ~6.3 GPa with a hardness value (H) of 0.86 GPa, which is surprisingly high considering that these films are stabilized by, and partly composed of, ~30 wt % NMP and water. This modulus value is also high when compared to values for other cross-linked, high-modulus polymers: Poly(imide)s typically have a Young's modulus value of 2.5 to 3.2 GPa, and values range between 2.0 and 4.0 GPa for nylon (Fig. 3B) (15). ODA PHTs were entirely stable in aqueous solutions, except in strongly acidic solutions (pH < 2), where they could be completely depolymerized, albeit at a slower rate than the HDCN (24 hours for 25.0 mg of ODA PHT versus 56 min for

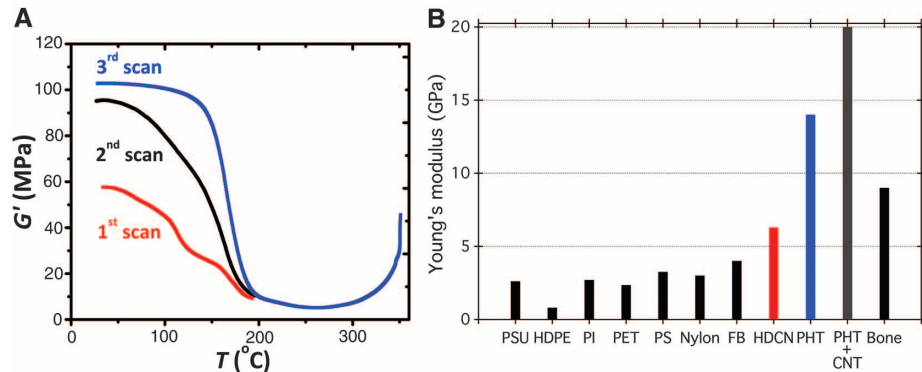


Fig. 3. Determination of modulus of materials. (A) The ODA HDCN was partially identified using dynamic mechanical analysis (DMA). Starting materials were heated at 50°C for 1 hour and cycled from 22° to 200°C. The first scan showed two thermal transitions, the T_g at ~125°C corresponding to the HDCN. By the third scan, a single T_g just under 200°C of the fully cured PHT was visible. (B) Nanoindentation measurements determined Young's modulus for ODA HDCN, PHT, and composite PHT materials (ODA-PHT Young's modulus was determined to be 10.3 GPa with Instron testing; see data in fig. S28). Mechanical properties of the PHT were enhanced through CNT addition (~20.0 GPa from ~14.0 GPa). This increased strength-to-weight ratio is commonly strived for in epoxy carbon fiber composites (19, 20). PSU, polysulfone; HDPE, high-density polyethylene; PI, polyimide; PET, poly(ethylene terephthalate); PS, polystyrene; FB, fiberboard.

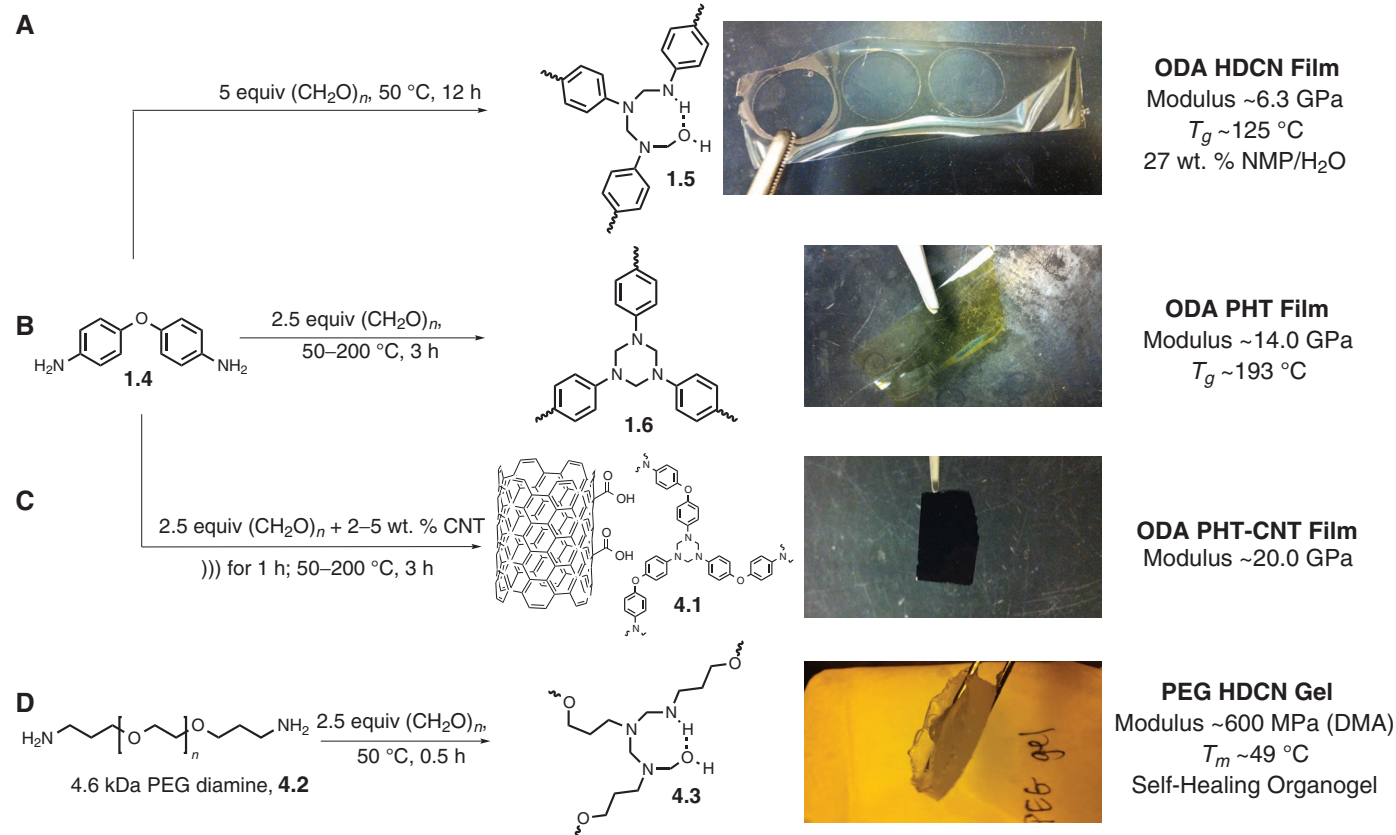


Fig. 4. Tunable HDCN/PHT platforms. (A) ODA HDCNs are cross-linked polymer networks plasticized with NMP and water that exhibit high modulus (~6.3 GPa), are flexible, and processable. The T_g was ~125°C and films decompose to monomer in strongly acidic solutions (pH <2) within an hour. (B) ODA PHTs exhibited a modulus of ~14.0 GPa, are robust materials, and are resistant to bases, oxidants, and

solvents. Rigid PHTs exhibited a T_g of ~192°C and decomposed at ~300°C. PHTs decompose to monomer after long exposure (>24 hours) to strong acid. (C) PHT film reinforced with 2 wt % of multiwalled, surface-oxidized CNTs exhibited a Young's modulus of ~20.0 GPa. (D) An HDCN derived from 4.6-kD PEG diamine formed a self-healing organogel, which was reversible in neutral water after 24 hours.

27.3 mg of ODA HDCN in 10 ml of 0.5 M H₂SO₄, pH = 0); PHTs were insensitive to weak acids, neutral water, and basic conditions (table S4).

The exceptional Young's modulus of aromatic PHTs, combined with the observed reversibility of the reaction in strongly acidic conditions, makes the PHTs an unprecedented new class of thermosets, as most high-modulus materials are completely chemically inert and often very difficult, if not impossible, to rework or recycle. Given their high stability toward organic solvents, ODA PHTs could be used in environments where robust, solvent-resistant materials are necessary; moreover, they resist catastrophic failure associated with environmental stress cracking (fig. S27). The acid-promoted reversibility of ODA PHTs would allow them to become the first thermosets that are easily repolymerized or recycled. The ability to depolymerize the fully cured PHTs opens numerous applications such as reworkable encapsulates (16), which have been explored extensively for recovery of high-value components such as in microelectronics, although approaches commonly sacrifice the mechanical properties of the resin. This rework can lead to substantial cost savings, as often in high-value applications, thermoset components with a single defect can result in the entire part being scrapped because the intractability of traditional thermosets prevents their removal to effect repairs.

PEG oligomers (4.2) were also studied for their reactivity with paraformaldehyde. Although PEG PHTs were not formed at high temperature, PEG HDCNs (4.3) that formed at low temperature exhibited properties that were markedly different from those of the high-modulus ODA HDCNs and PHTs. HDCNs based on oligomeric PEG diamines formed elastic organogels with a melting temperature (T_m) of ~49°C by DSC, that com-

pletely reverted back to their original starting components in neutral water, which makes them promising candidates for reversible covalent constructs for a wide variety of applications that require reversible assemblies such as materials for cargo delivery. In addition to reversibility, PEG HDCN organogels exhibited self-healing properties; images and a video of its self-healing ability can be found in the supplementary materials (movie S1 and fig. S23). Previously reported polymerizations attempting to form PHTs focused on low-temperature reactions (50° to ~100°C) and used soluble aliphatic diamine monomers, or targeted PHTs, as reactive intermediates (17). These intermediate aliphatic diamine-based homopolymers were not extensively considered as useful thermosets because they exhibited poor mechanical properties (18). We believe, given the high rate of the reaction to its gelling point (within minutes) and difficulties we encountered to spectroscopically characterize ODA PHTs and HDCNs, that at these lower temperatures, low cross-link density, low-modulus HDCNs were most likely formed.

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Supplementary Materials

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Marine Ice Sheet Collapse Potentially Under Way for the Thwaites Glacier Basin, West Antarctica

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Resting atop a deep marine basin, the West Antarctic Ice Sheet has long been considered prone to instability. Using a numerical model, we investigated the sensitivity of Thwaites Glacier to ocean melt and whether its unstable retreat is already under way. Our model reproduces observed losses when forced with ocean melt comparable to estimates. Simulated losses are moderate (<0.25 mm per year at sea level) over the 21st century but generally increase thereafter. Except possibly for the lowest-melt scenario, the simulations indicate that early-stage collapse has begun. Less certain is the time scale, with the onset of rapid (>1 mm per year of sea-level rise) collapse in the different simulations within the range of 200 to 900 years.

Glaciers along the Amundsen Coast of Antarctica are thinning (1, 2), producing the majority of Antarctica's contribution to sea-level rise (3, 4). Much of this thinning is probably a response to the increased presence of warm modified Circumpolar Deep Water (CDW)

on the adjacent continental shelf (5, 6), which is melting and thinning the floating ice shelves that buttress the ice sheet (7–9). Thinner ice shelves are less able to restrain flow from the interior, contributing to feedbacks that increase ice discharge to the ocean (10–14). Thwaites and Haynes

Glaciers, which hereafter we refer to collectively as Thwaites Glacier, produce just under half (52 Gt/year in 2007) of the Amundsen Coast losses (105 Gt/year in 2007) (3, 4, 15, 16), making it one of the largest contributors to sea-level change. This glacier and the immediately adjacent and rapidly thinning Pine Island Glacier (2, 3) were identified as potentially unstable several decades ago (17).

The present Thwaites grounding line—the location where ice reaches the ocean and goes afloat—rests on a coastal sill ~600 m below sea level (bsl) (Fig. 1) (18). At ~60 to 80 km farther inland, this sill gives way to a deep (>1200 m bsl) marine basin, yielding the potential for marine ice-sheet instability (13, 17, 19–21). Ice discharge is nonlinearly proportional to grounding-line thickness. Hence the potential for instability exists where the ice-sheet bed lies below sea level and steepens toward the interior, so that an initial retreat into deeper water creates a feedback,

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